VARDHAMAN
COLLEGE OF ENGINEERING

A COURSE END PROJECT REPORT ON

MOVIE RATING AND REVENUE ANALYSER

A9513 – Python Programming Laboratory

Submitted by

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VARDHAMAN COLLEGE OF ENGINEERING

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This is to certify that the Course End Project titled **"MOVIE RATING AND REVENUE ANALYSER"** is carried out by **U ADARSH , D ARUNA and T YASHASVIN REDDY** with Roll Numbers **26885A0557, 26885A0558 , 25881A05KX** towards **A9513 – Python Programming Laboratory** course in partial fulfilment of the requirements for the award of degree of **Bachelor of Technology** in **Computer Science and Engineering** during the Academic year **2026-27**.

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Abstract

Movies are evaluated using several factors, including audience ratings, number of votes, and revenue earned. These values are useful for understanding audience response and commercial performance. The Movie Rating and Revenue Analyser is a simple data-analysis project that organises movie information and presents meaningful summaries so that users can compare movies without manually inspecting every record.

The proposed system uses Python to load a movie dataset from a CSV file, validate and clean the records, calculate rating and revenue statistics, identify high-rated and high-revenue movies, and display results through tables and charts. The project is designed as an educational prototype. Its results depend on the dataset supplied by the user; ratings and revenue are not treated as proof that one factor causes another. The analyser can be extended with a graphical interface, more data sources, and predictive models.

Keywords: Movie Analytics, Python, Pandas, Data Cleaning, Ratings, Revenue, Data Visualisation, CSV Datase

1 Introduction

The film industry produces a large amount of information about movies, such as title, release year, genre, audience rating, number of votes, and box-office revenue. When the information is stored in a spreadsheet or a dataset containing many rows, comparing movies manually can be slow and may lead to mistakes. A computer-based analyser can process these records consistently and produce useful summaries.

The Movie Rating and Revenue Analyser is a software application that reads movie records and analyses two important measures: rating and revenue. Rating represents a score given by viewers or a rating service, while revenue represents the money reported as earned by a movie. These measures describe different aspects of performance. A movie may receive a high rating but earn modest revenue, or earn high revenue while receiving an average rating.

The project applies basic concepts of programming and data analysis. A dataset is represented as rows and columns. Each row describes a movie, and each column represents an attribute. The application checks missing values and data types before performing calculations. For ratings, it can calculate average rating and sort movies by rating. For revenue, it can calculate total and average revenue and rank movies by revenue when the data is available.

Data visualisation is used to make results easier to understand. Bar charts can compare the top movies by rating or revenue, while summary tables show the underlying values. The project does not assume that ratings directly cause revenue; it presents descriptive statistics that help users explore patterns. The final output is intended for students, movie enthusiasts, and anyone learning basic data analytics.

Problem Statement

To develop a simple and understandable program that processes movie data and provides reliable summaries of ratings and revenue, reducing the effort required for manual comparison.

Scope of the Project

- Reads movie information from a structured CSV dataset.
- Cleans basic data issues such as blank titles, missing values, and invalid numeric entries.
- Calculates rating and revenue summaries and displays ranked movie lists.

2 Literature Review

Data analysis systems commonly follow a workflow of data collection, cleaning, transformation, analysis, and presentation. Structured files such as CSV are frequently used in introductory projects because they are portable and can be inspected with ordinary spreadsheet software. Python libraries provide tools to automate these steps and make the analysis repeatable.

Pandas is a Python library used for working with tabular data. It supports loading CSV files, selecting columns, handling missing values, sorting records, and calculating statistics such as mean, count, minimum, and maximum. These features are suitable for processing movie records and producing consistent summaries. Data cleaning is important because missing or incorrectly formatted revenue and rating values can distort calculations.

Matplotlib is a widely used plotting library that can present results as bar charts, histograms, and other visual forms. In a movie-analysis application, charts make it easier to compare the highest-rated movies and movies with the largest reported revenue. However, visual results should be interpreted carefully: a ranking can be affected by sample size, missing data, currency units, and differences in the period or market covered by the dataset.

The proposed project combines these basic methods in one small workflow. Unlike a production analytics platform, it does not automatically retrieve live box-office figures or guarantee that all movies are represented. Its main contribution is an easy-to-understand implementation that demonstrates how a structured dataset can be converted into readable summaries and visual outputs.

3 Objectives

- To load and prepare movie data from a CSV file by checking columns, missing values, and numeric fields.
- To analyse movie ratings and revenue using summary statistics and ranking operations.
- To present the analysis through readable tables and charts so that movie performance can be compared easily.

4 Methodology

4.1 Architectural Diagram

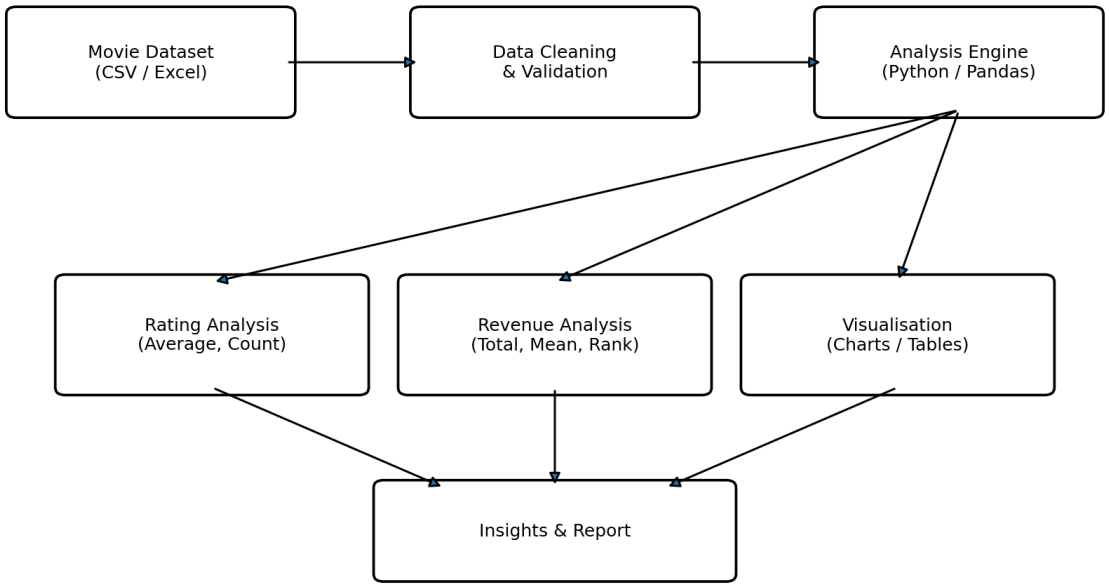


Figure 1: Architecture of the Movie Rating and Revenue Analyser

The system begins with a movie dataset supplied as a CSV file. The data-preparation stage checks the required columns and converts rating and revenue fields into numeric values. The analysis engine calculates summaries and ranks records. The rating and revenue modules produce tables, while the visualisation module creates charts. The final output consists of summaries and insights that can be used for comparison.

4.2 Hardware and Software Requirements

Category	Requirement
Hardware	Laptop or desktop computer; 4 GB RAM or higher recommended; sufficient storage for the dataset.
Operating System	Windows, Linux, or macOS.
Programming Language	Python 3.x.
Libraries	Pandas for tabular data; Matplotlib for charts; NumPy may be used for numerical operations.
Input	CSV file containing movie title, rating, and revenue columns; genre/year may be included optionally.
Output	Summary tables, ranked movie lists, and charts displayed or saved as image files.
Development Tool	IDLE, VS Code, PyCharm, Jupyter Notebook, or another Python-compatible editor.

4.3 Working Principle

The working principle is based on a sequence of data-processing steps. First, the program reads the dataset and confirms that the required columns exist. The title field is cleaned by removing unnecessary spaces, and rating and revenue fields are converted to numeric form. Rows with missing or invalid values are handled according to the analysis being performed, so a missing revenue does not have to remove an otherwise usable rating.

1. Input: The user places the movie dataset in CSV format and provides the file path.
2. Validation: The program checks for required columns such as title, rating, and revenue.
3. Cleaning: Empty titles and invalid numeric entries are identified; missing values are handled transparently.
4. Analysis: The program calculates average rating, number of records, total revenue, and average revenue where values are available.
5. Ranking: Movies are sorted by rating or revenue to identify the leading records.
6. Visualisation: Bar charts display selected high-rated and high-revenue movies.
7. Output: The program prints summary statistics and presents charts for interpretation.

The average rating is calculated by dividing the sum of valid ratings by the number of valid rating values. Average revenue is calculated similarly using valid revenue values. Total revenue is the sum of valid revenue entries. These calculations are only meaningful when the dataset uses consistent units and comparable definitions.

4.1 Software Implementation

Algorithm

- Start the program.
- Import Pandas and Matplotlib.
- Read the movie CSV file into a data table.
- Check whether the title, rating, and revenue columns are available.
- Clean titles and convert rating and revenue to numeric values.
- Calculate the average rating, average revenue, and total revenue from valid entries.
- Sort the dataset to obtain the top movies by rating and by revenue.
- Display summary tables and generate comparison charts.
- Stop the program.

C source code

```
import pandas as pd
import matplotlib.pyplot as plt

# CSV should contain: title, rating, revenue
file_name = "movies.csv"
df = pd.read_csv(file_name)

# Standardise column names and validate required fields
df.columns = [col.strip().lower() for col in df.columns]
required = {"title", "rating", "revenue"}
missing = required - set(df.columns)
if missing:
    raise ValueError(f"Missing required columns: {missing}")

# Clean and convert fields
df["title"] = df["title"].astype(str).str.strip()
df["rating"] = pd.to_numeric(df["rating"], errors="coerce")
df["revenue"] = pd.to_numeric(df["revenue"], errors="coerce")
df = df[df["title"].ne("") & df["title"].ne("nan")]

# Summaries
```

```
print("Number of movies:", len(df))
print("Average rating:", round(df["rating"].mean(), 2))
print("Total reported revenue:", round(df["revenue"].sum(), 2))
print("Average reported revenue:", round(df["revenue"].mean(), 2))

# Top 10 movies by rating and revenue
top_rated = df.dropna(subset=["rating"]).sort_values(
    "rating", ascending=False
).head(10)
top_revenue = df.dropna(subset=["revenue"]).sort_values(
    "revenue", ascending=False
).head(10)

print("\nTop-rated movies:")
print(top_rated[["title", "rating"]].to_string(index=False))
print("\nTop-revenue movies:")
print(top_revenue[["title", "revenue"]].to_string(index=False))

# Charts
if not top_rated.empty:
    top_rated.plot.bar(x="title", y="rating", legend=False,
                       title="Top Movies by Rating")
    plt.ylabel("Rating")
    plt.tight_layout()
    plt.show()

if not top_revenue.empty:
    top_revenue.plot.bar(x="title", y="revenue", legend=False,
                         title="Top Movies by Revenue")
    plt.ylabel("Revenue (dataset units)")
    plt.tight_layout()
    plt.show()
```

Code explanation: Pandas loads the dataset and performs data cleaning and statistical calculations. The required-column check helps detect an incorrectly formatted input file early. The program treats invalid rating and revenue values as missing rather than allowing them to affect calculations. The two ranking operations are independent, because the movie with the highest rating may not have the highest revenue. Matplotlib displays the resulting comparisons.

Expected Dataset Format

title	rating	revenue
Sample Movie A	8.2	1250000
Sample Movie B	7.6	2400000
Sample Movie C	8.8	980000

The values above are fictional examples used only to demonstrate the expected file structure. They are not real box-office results. Revenue should be recorded in one consistent unit, such as rupees or US dollars, and the unit should be documented in the final dataset.

5 Results and Discussion

When the program is run with a correctly formatted CSV file, it reports the number of movie records, average rating, total reported revenue, and average reported revenue. It also displays two ranked lists: movies with the highest available ratings and movies with the highest available revenue. Charts make these rankings easier to compare visually.

Expected Output

Output item	Description
Record count	Number of movie rows retained after title cleaning.
Average rating	Mean of valid rating values; rounded to two decimal places.
Total revenue	Sum of valid revenue values in the dataset's units.
Average revenue	Mean of valid revenue values.
Top-rated list	Up to ten movies ordered by rating, highest first.
Top-revenue list	Up to ten movies ordered by reported revenue, highest first.
Charts	Bar charts for the top-rated and top-revenue movies.

Discussion

The analyser helps distinguish audience response from commercial performance. Rating-based ranking shows which titles have the highest reported scores, whereas revenue-based ranking shows which titles have the largest reported earnings. Comparing these lists can reveal that the two rankings are not necessarily identical. The output is descriptive and does not establish why a movie earned a particular amount.

The accuracy of the results depends on the dataset. Missing ratings reduce the number of movies that can be ranked by rating, while missing revenue values reduce the number of records used for revenue statistics. If revenue is reported in different currencies or for different time periods, comparisons may be misleading. Therefore, the data source, currency, and coverage period should be recorded.

Achievement of Objectives

- The project provides a structured way to load and clean movie records.
- It calculates basic rating and revenue statistics and ranks movies using the available values.
- It presents results as tables and charts, making comparisons easier than manual inspection.

Limitations and Future Improvements

- The current version requires a prepared CSV file and does not automatically fetch live data.
- A graphical user interface can be added to let users choose files and filters.
- More attributes such as genre, release year, budget, votes, and language can be included.
- Data validation can be strengthened with source tracking, currency conversion, and duplicate detection.
- Future work may explore correlations between budget, rating, votes, and revenue, while avoiding claims of causation from correlation alone.

6 Mapping POs and SDGs

6.1 Program Outcomes (POs) Mapping

The following program outcomes are relevant to the development and evaluation of the Movie Rating and Revenue Analyser.

PO No.	Program Outcome	Relevance to Project
PO1	Engineering Knowledge	Uses programming, basic statistics, and data-processing concepts to analyse movie records.
PO2	Problem Analysis	Identifies issues in manual comparison and examines missing or invalid dataset values.
PO3	Design/Development of Solutions	Designs a workflow that converts raw records into summaries, rankings, and charts.
PO4	Conduct Investigations of Complex Problems	Explores rating and revenue patterns through data inspection and comparison.
PO5	Engineering Tool Usage	Uses Python, Pandas, and Matplotlib to process and visualise data.
PO8	Individual and Collaborative Teamwork	Can be developed collaboratively by dividing data preparation, coding, testing, and report tasks.
PO9	Communication	Communicates findings through readable tables, plots, and a written report.
PO10	Project Management and Finance	Requires planning of tasks and consideration of revenue as a financial measure in the dataset.
PO12	Life-Long Learning	Provides a foundation for learning further data analytics and visualisation techniques.

Table 1: Mapping of Program Outcomes (POs) to Project Relevance

6.2 Sustainable Development Goals (SDGs) Mapping

This is an educational data-analysis project and has an indirect relationship with the Sustainable Development Goals. The most relevant mapping is quality education through practical programming and data-literacy skills.

SDG No.	Goal	Relevance to Project
SDG 4	Quality Education	Supports practical learning in programming, statistics, data cleaning, and visualisation.
SDG 9	Industry, Innovation and Infrastructure	Demonstrates a small digital tool that processes information to support analysis and decision-making.

Table 2: Mapping of Sustainable Development Goals (SDGs) to Project Relevance

7. CONCLUSION

The Movie Rating and Revenue Analyser demonstrates how Python can be used to turn a structured movie dataset into useful summaries. The system loads movie records, checks important fields, cleans numeric values, calculates rating and revenue statistics, ranks movies, and presents comparisons using tables and charts. It provides a practical example of applying programming and basic data analytics to a familiar domain.

The project meets its stated objectives at prototype level. It reduces manual effort and makes it easier to compare audience ratings with reported revenue, while keeping the two measures separate. The results should be interpreted according to the quality and limitations of the input data. Future development can include a user interface, more attributes, improved data validation, and reliable data-source integration. With these improvements, the project can become a more flexible tool for exploring movie-related data.

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- Python Software Foundation, “The Python Tutorial.” https://docs.python.org/3/tutorial/
- Project dataset: Movie records provided by the project team. Record the actual dataset name, source URL, access date, currency, and licence here before final submission.
- Note: The implementation and sample dataset shown in this report are a proposed academic prototype. Before submission, run the program with the actual project dataset, add screenshots of the real output, and update the student name, roll number, and batch details.